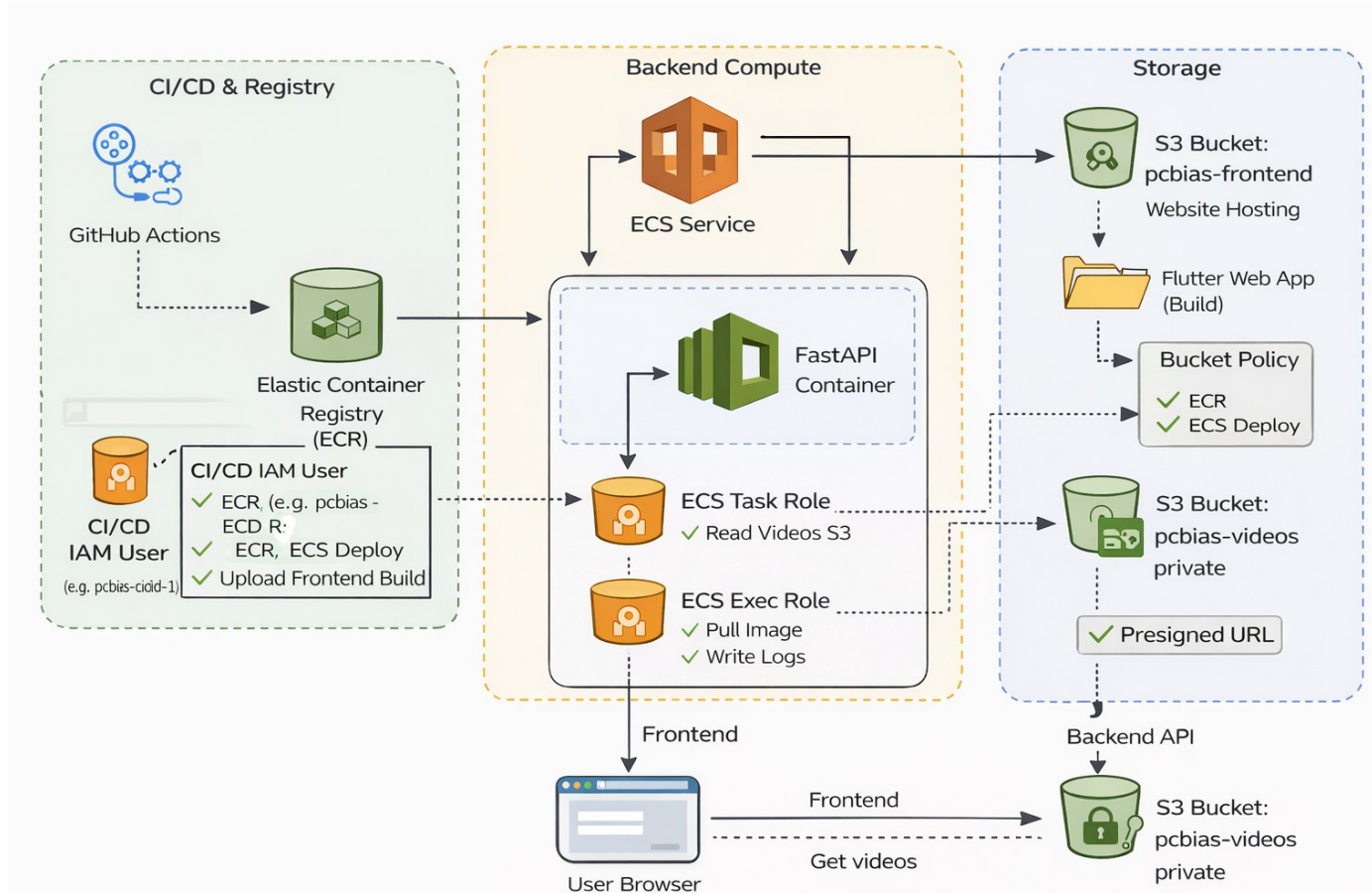


Cloud Deployment Architecture, now with IaC



Cloud formation stacks

☰

CloudFormation > Stacks

CloudFormation <

Stacks

Stack refactors [New](#)

StackSets

Exports

Infrastructure Composer

IaC generator

Hooks overview

Invocation summary

Hooks

▼ Registry

Public extensions

Activated extensions

Publisher

Stacks (3)

🔄

Delete stack

Update stack ▼

Stack actions ▼

Create stack ▼

Filter status

🔍 Search by stack name

Active ▼

☒ View nested

< 1 > ⚙️

	Stack name	Status	Created time ▼	Description
☐	inspection-api-backend-resources	✅ CREATE_COMPLETE	2026-03-07 16:04:10 UTC+0530	Backend Infrastructure Stack for PCBias Inspection App
☐	inspection-api-network-resources	✅ CREATE_COMPLETE	2026-03-07 15:57:33 UTC+0530	Network Stack for PCBias Inspection App
☐	inspection-api-shared-resources	✅ CREATE_COMPLETE	2026-03-07 15:54:28 UTC+0530	Shared Resources Stack for PCBias Inspection App

CloudShell

Feedback

📱 Console mobile app

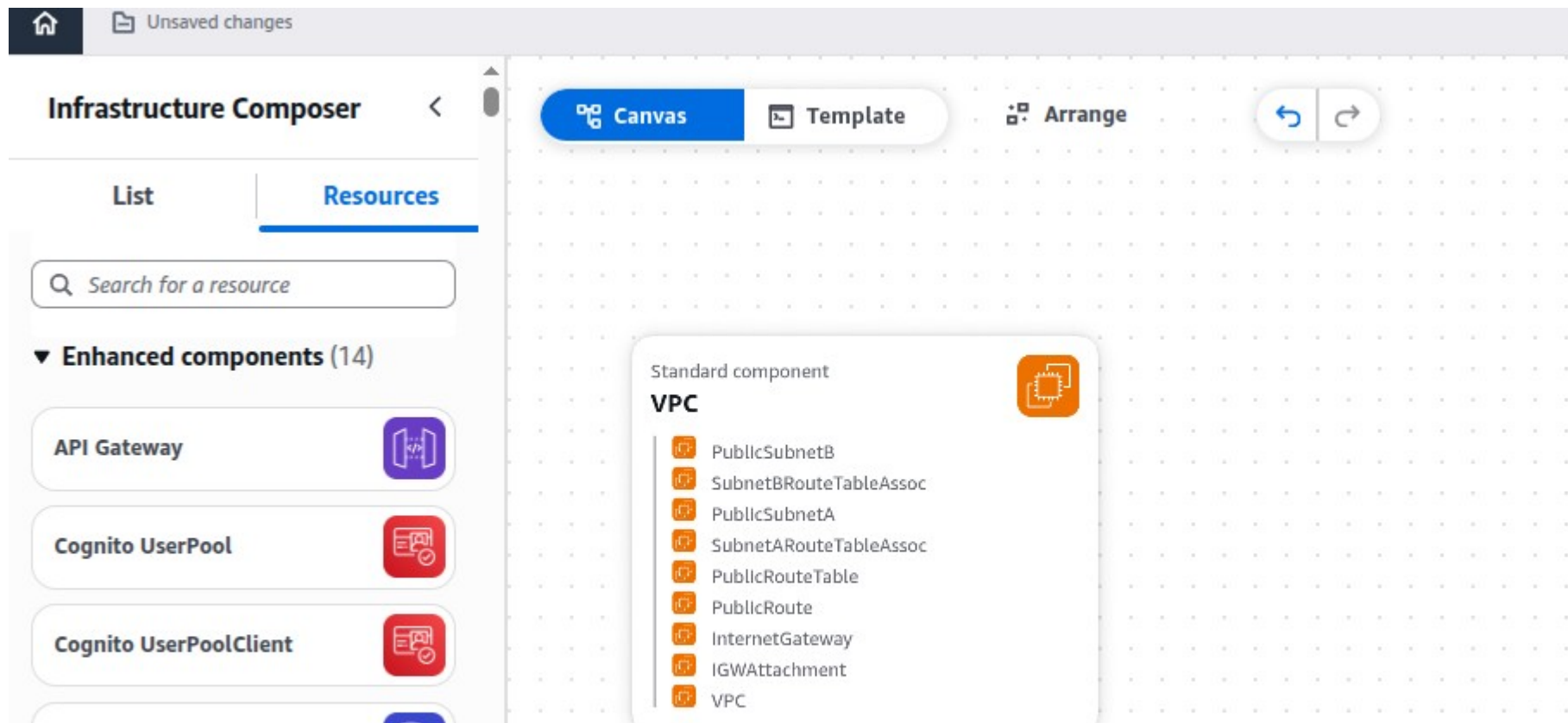
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Privacy

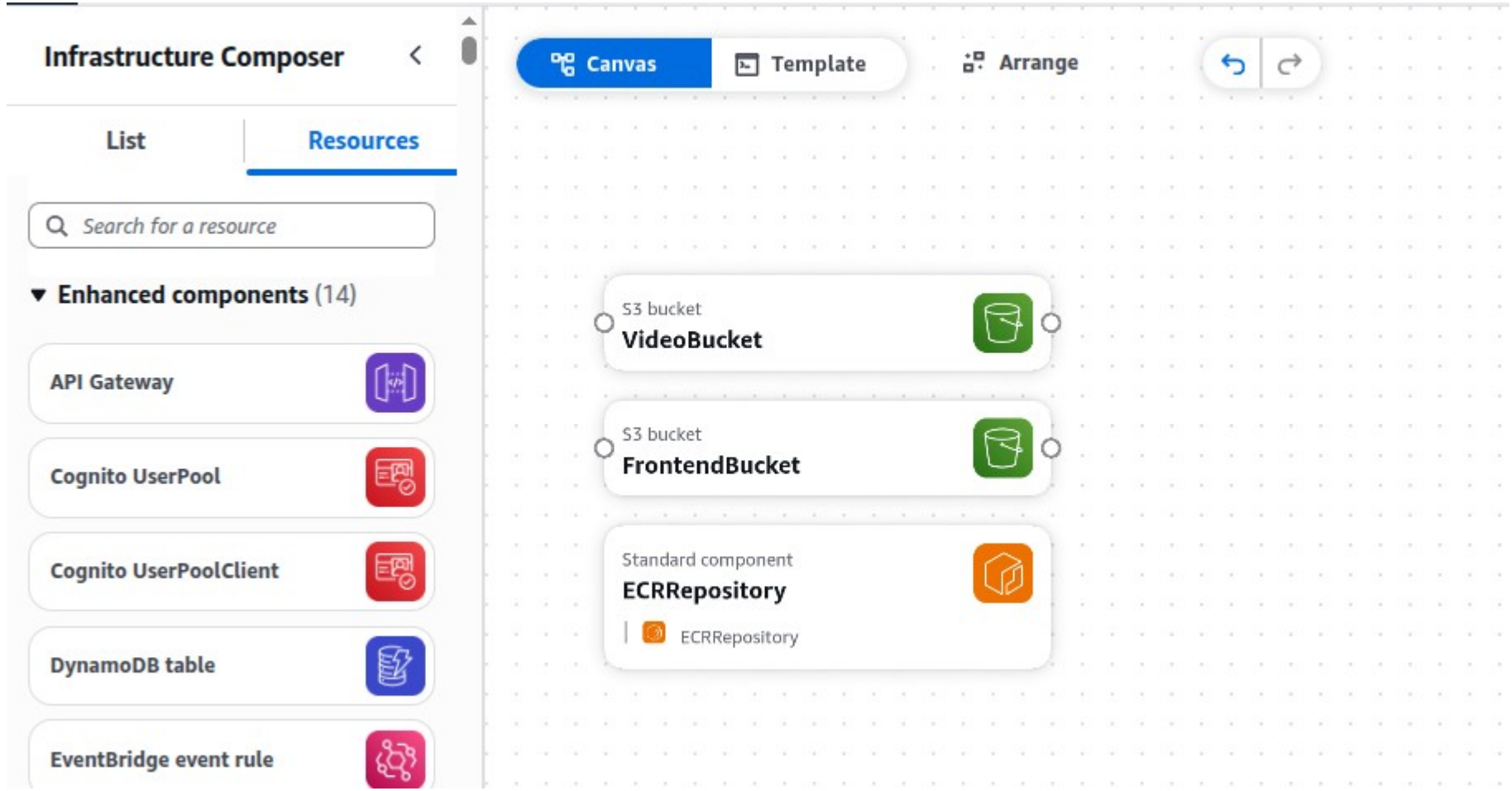
Terms

Cookie preferences

The Network Stack



The Shared Resources Stack



The screenshot displays the AWS CloudFormation console interface for an **Infrastructure Composer** stack. The left sidebar shows the **Resources** tab, listing enhanced components (14) including API Gateway, Cognito UserPool, Cognito UserPoolClient, DynamoDB table, and EventBridge event rule. The main canvas displays the stack's resources:

- VideoBucket**: S3 bucket
- FrontendBucket**: S3 bucket
- ECRRepository**: Standard component

The canvas also shows the **ECRRepository** resource type icon and label.

The Backend Stack

Search for a resource

Enhanced components (14)

API Gateway



Cognito UserPool



Cognito UserPoolClient



DynamoDB table



EventBridge event rule



EventBridge schedule



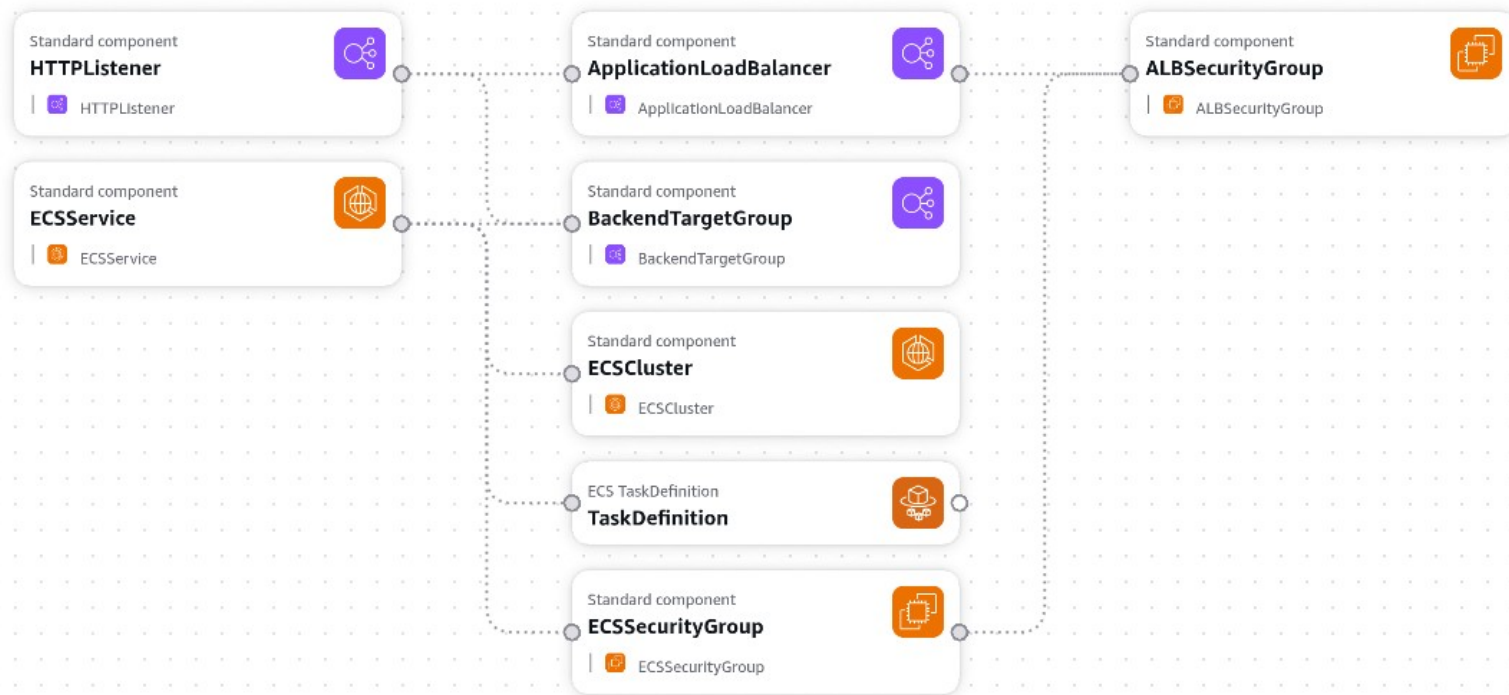
Kinesis stream



Lambda function



Lambda layer



Real time monitoring of infrastructure provisioning

CloudFormation > Stacks > inspection-api



Stacks (1)



Filter status

Search by stack name

Active

View nested

< 1 > ⚙

Stacks

inspection-api

2026-03-06 17:40:52 UTC+0530

CREATE_IN_PROGRESS

inspection-api

Delete stack

Update stack

Stack actions

Create stack

Stack info

Events

Resources

Outputs

Parameters

Template

Changesets

Git sync

Resources (17)

Search resources

< 1 > ⚙

Logical ID	Physical ID	Type	Status	Module
ALBSecurityGroup	sg-032ac0cdcac28b1c	AWS::EC2::SecurityGroup	CREATE_IN_PROGRESS	-
BackendTargetGroup	arn:aws:elasticloadbalancing:eu-north-1:586177432842:targetgroup/pcbiasm-tg/731d614f72870abc	AWS::ElasticLoadBalancingV2::TargetGroup	CREATE_IN_PROGRESS	-
ECRRepository	pcbiasm-inspection-api	AWS::ECR::Repository	CREATE_COMPLETE	-
ECSCluster	pcbiasm-cluster	AWS::ECS::Cluster	CREATE_COMPLETE	-
ECSExecutionRole	pcbiasm-ecs-execution-role	AWS::IAM::Role	CREATE_IN_PROGRESS	-
ECSTaskRole	-	AWS::IAM::Role	CREATE_IN_PROGRESS	-
FrontendBucket	pcbiasm-frontend	AWS::S3::Bucket	CREATE_COMPLETE	-

The Result?

Automated Deployment!

It worked!

← Deploy Backend to ECS

✓ **final final change ci cd #16** Re-run all jobs 🕒 Latest #3 ▾ ⋮

🏠 Summary

All jobs ☰

✓ **Provision, Build, Push and Deploy**

Run details

🕒 Usage

📄 Workflow file

Provision, Build, Push and Deploy

succeeded 25 minutes ago in 4m 53s

🔍 Search logs 🔄 ⚙️

> ✓ Set up job	1s
> ✓ Checkout repository	1s
> ✓ Configure AWS credentials	0s
> ✓ Collect Stack Outputs	4s
> ✓ Deploy backend stack	3m 8s
> ✓ Get ECS cluster name	0s
> ✓ Login to Amazon ECR	1s
> ✓ Build and push Docker image	1m 35s

It worked!

✓ final frontend ci cd #7

Re-run all jobs

...

Summary

All jobs

build-and-deploy

Run details

Usage

Workflow file

build-and-deploy

succeeded 16 minutes ago in 1m 34s

Search logs

⚙

> ✓ Set up job

2s

> ✓ Checkout code

1s

> ✓ Configure AWS credentials

1s

> ✓ Get ALB URL from CloudFormation

6s

> ✓ Set up Flutter

26s

> ✓ Install dependencies

1s

> ✓ Build Web Release

48s

> ✓ Get frontend bucket name

2s

> ✓ Sync to S3

3s

> ✓ Post Set up Flutter

0s

> ✓ Post Configure AWS credentials

0s

Elastic Container Registry

Amazon Elastic Container Service

▼ Private registry

Repositories

Features & Settings

▼ Public registry

Repositories

Settings

Amazon ECR > Private registry > Repositories

Private repositories (1)

View push commands

Delete

Actions ▼

Create repository

Search by repository substring

	Repository name ▲	URI	Created at ▼	Tag immutability	Encryption type
☐	inspection-api	Confidential	04 March 2026, 13:40:19 (UTC+05.5)	Mutable	AES-256

Elastic Container Service

Amazon Elastic Container Service

Express Mode

Clusters

Namespaces

Task definitions

Account settings

Amazon ECR

Repositories

AWS Batch

Online learning workshop

Documentation

Discover products

Subscriptions

Amazon Elastic Container Service > Clusters > pcbias-cluster > Services

Cluster overview

ARN
arn:aws:ecs:us-east-1:586177432:842:cluster/pcbias-cluster

Status
Active

CloudWatch monitoring
Default

Registered container instances
-

Services

Tasks

Draining
-

Active
1

Pending
-

Running
1

Services

Tasks

Infrastructure

Metrics

Scheduled tasks

Configuration

Event history

Tags

Services (1) Info

Last updated 7 March 2026, 16:26 (UTC+5:30)

Manage tags

Update

Delete service

Create

Filter services by value

Filter launch type
Any launch type

Filter scheduling strategy
Any scheduling strategy

Filter resource management type
Any resource management type

< 1 >

	Service name	ARN	Status	Schedu...	L...	Task de...	Deployments and tasks
☐	pcbias-backend	arn:aws:ecs:us-e	Active	REPLICA	FAR...	pcbias-tas...	1/1 task

CloudShell

Feedback

Console mobile app

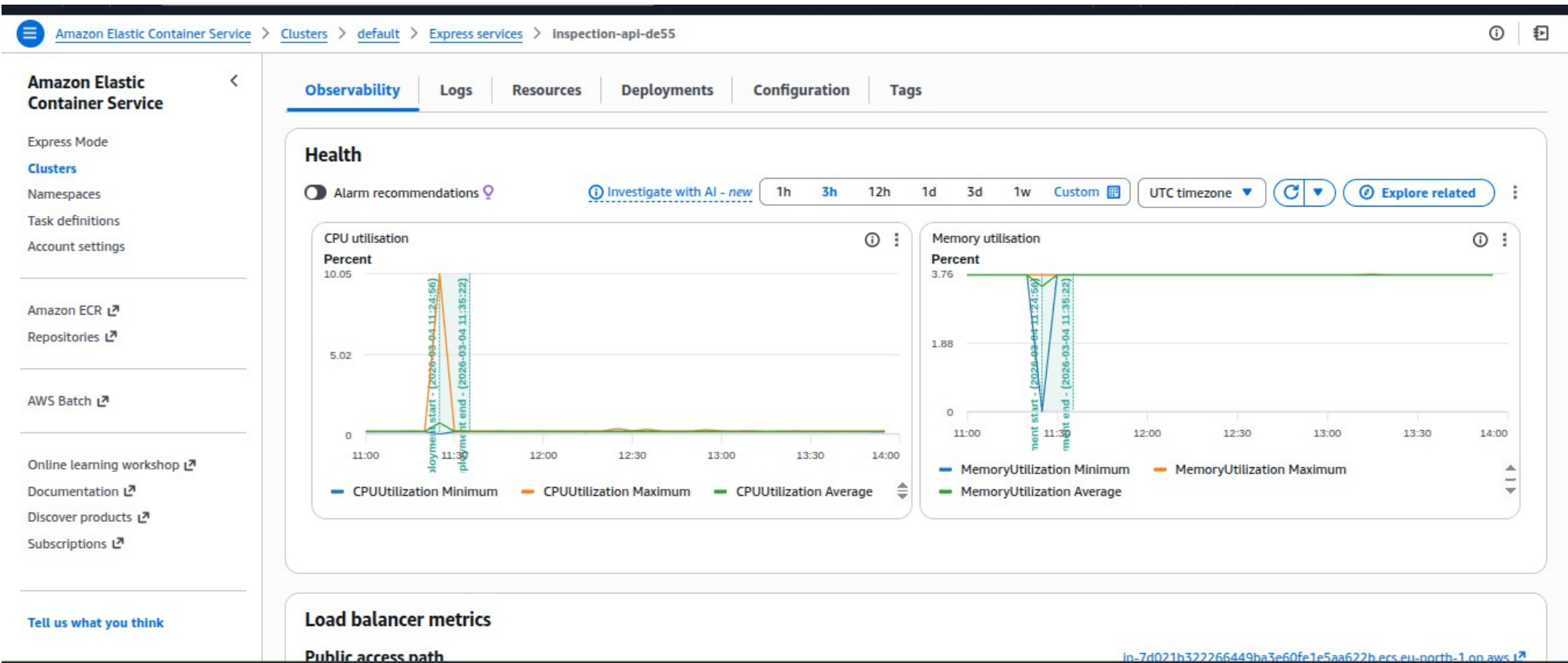
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Elastic Container Service



IAM Permissions Used by Github Actions

The screenshot shows the AWS IAM console interface. The left sidebar contains navigation links: Dashboard, Access Management (expanded), User groups, Users (selected), Roles, Policies, Identity providers, Account settings, and Root access management. The main content area is titled 'Permissions policies (6)' and shows a list of policies attached to the user 'pcbias-cicd-1'. The policies are all AWS managed and attached via the 'pcbias' group, except for 'AmazonECS_FullAccess' which is attached directly. The table includes columns for checkboxes, policy names, types, and attachment methods.

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access Management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings
- Root access management

Permissions policies (6)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: All types

☐	Policy name ↗	Type	Attached via ↗
☐	AmazonEC2ContainerRegistryFullAccess	AWS managed	Group pcbias
☐	AmazonECS_FullAccess	AWS managed	Directly
☐	AmazonElasticContainerRegistryFullAccess	AWS managed	Group pcbias
☐	AmazonS3FullAccess	AWS managed	Group pcbias
☐	CloudFrontFullAccess	AWS managed	Group pcbias
☐	IAMReadOnlyAccess	AWS managed	Group pcbias

Thank You